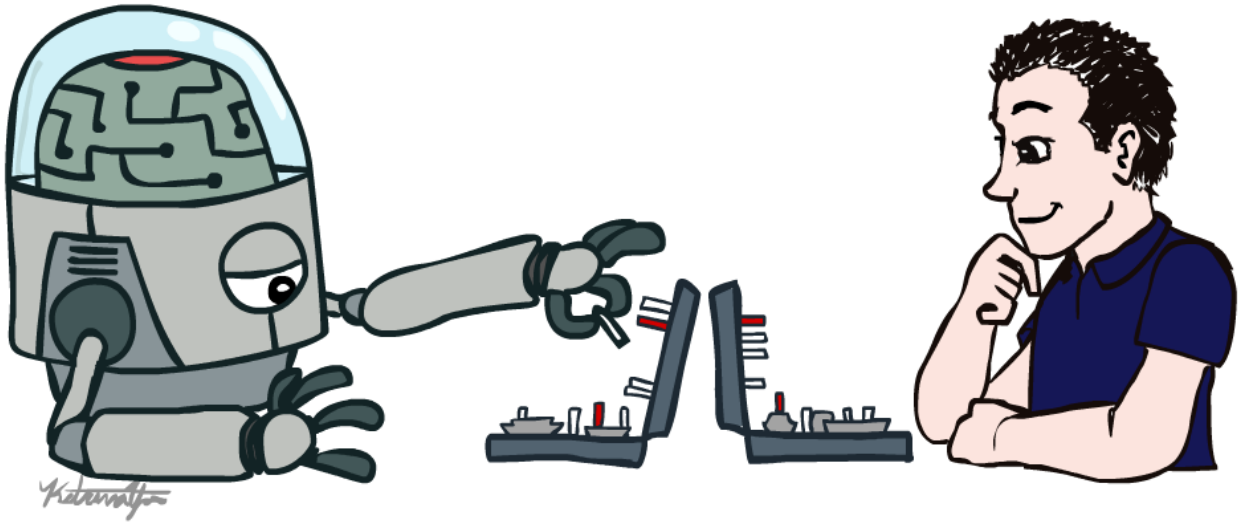
ICSI263: Машин сургалт

Linear Regression (1)

Getting started with linear regression



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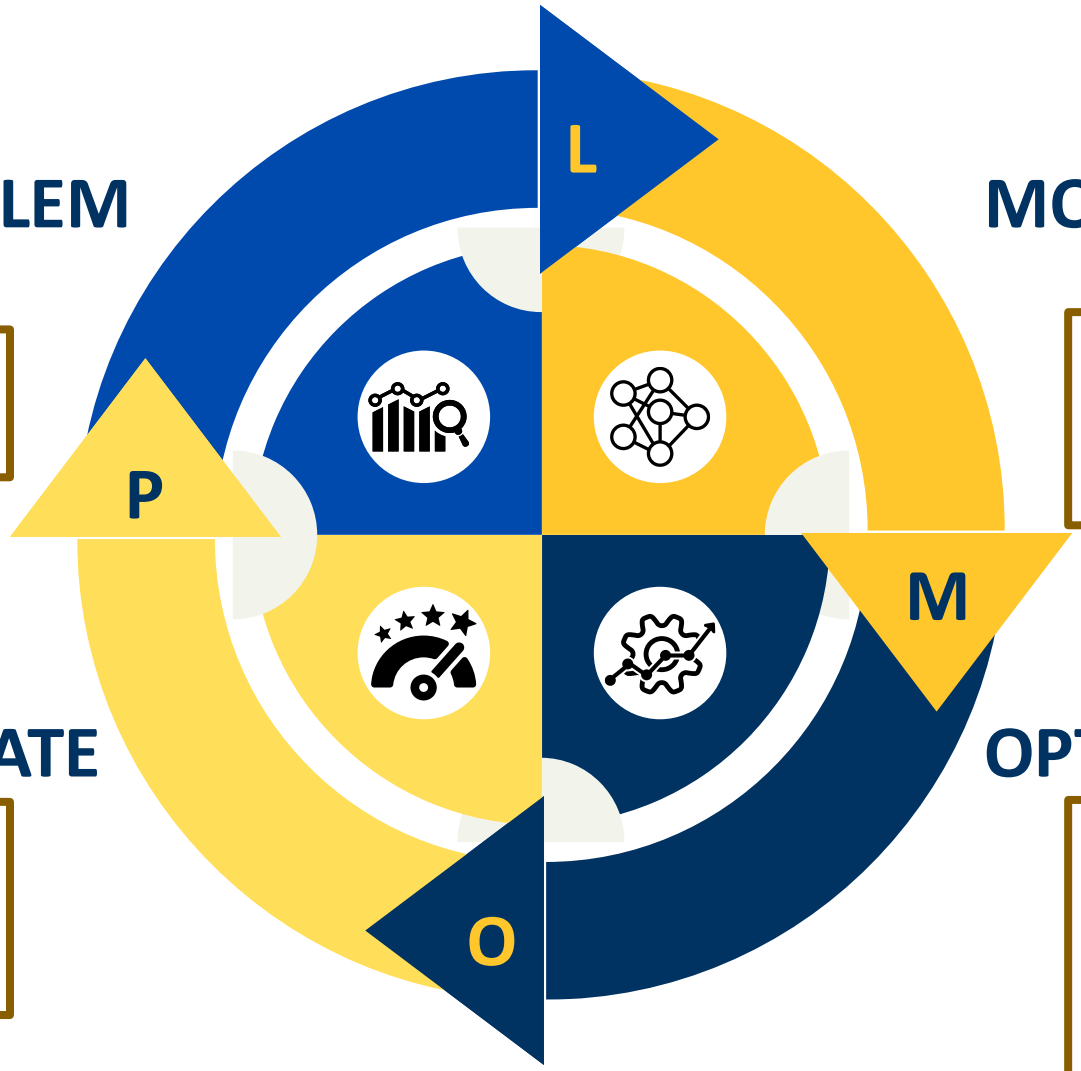
Roadmap

- Linear Regression Formulation
- Basis Functions
- Vectorizing Calculations
- Error Function
- Error Function Minimization
- Geometric Interpretation
- Evaluation

Linear Regression

- Linear Regression
- Basis Functions
- Vectorizing Calculations
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- Geometric Interpretation
- Evaluation

Linear Regression Outline



LEARNING PROBLEM

Supervised learning of scalar target values

MODEL DESIGN

Linear Regression with basis functions

$$y(\vec{x}, \vec{w}) = x^T \vec{w}$$

PREDICT & EVALUATE

Evaluate Predictions

$$\hat{y} = x_{test}^T w^*$$

on test set

OPTIMIZATION

Derive direct solution

$$E(w) = \frac{1}{2} \sum_{n=1}^{N} \left(t_n - y(x_n, w) \right)^2$$

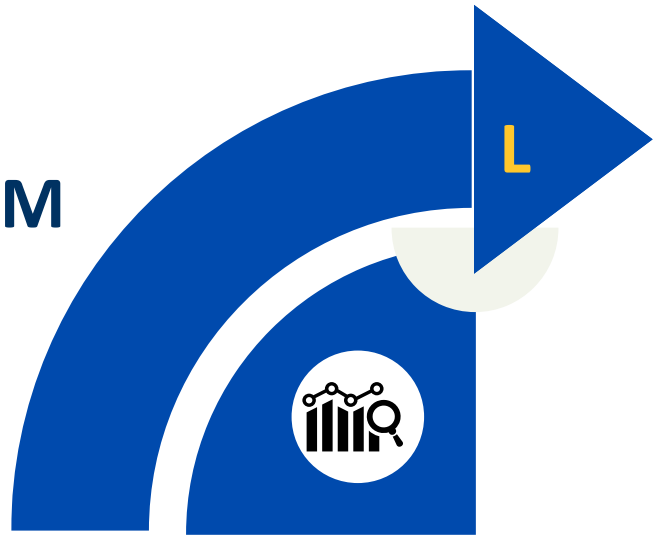
$$w^* = \underset{w}{\operatorname{argmin}}\, E(w)$$

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Learning Problem

LEARNING PROBLEM

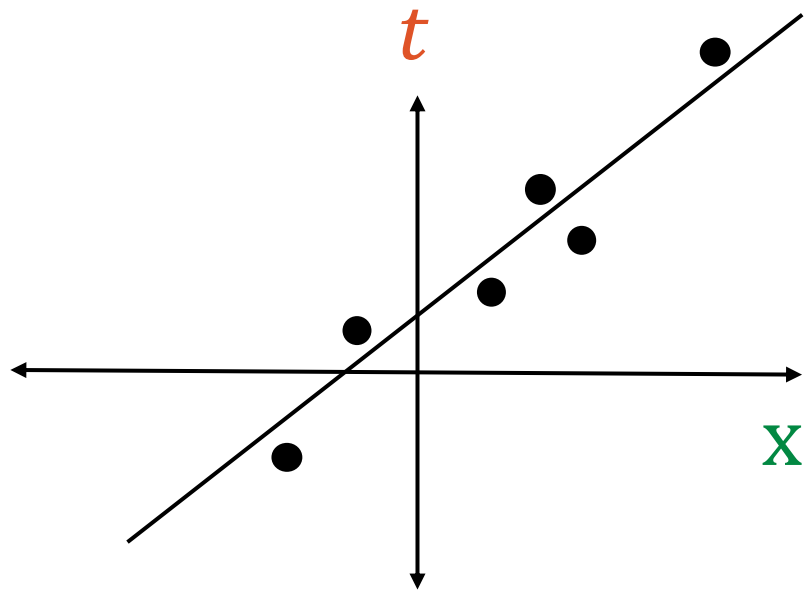
Supervised learning
of scalar target values

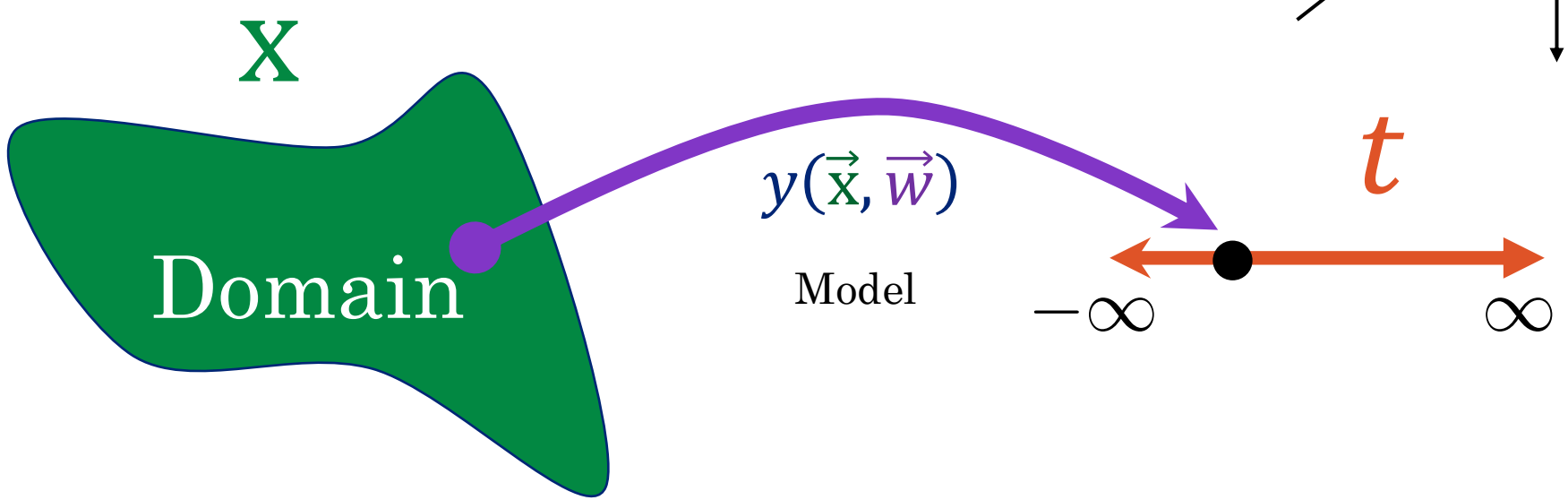


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Regression

- Estimating relationship between x and t.
  - t is a quantitative value
  - We will soon see x can be almost anything …



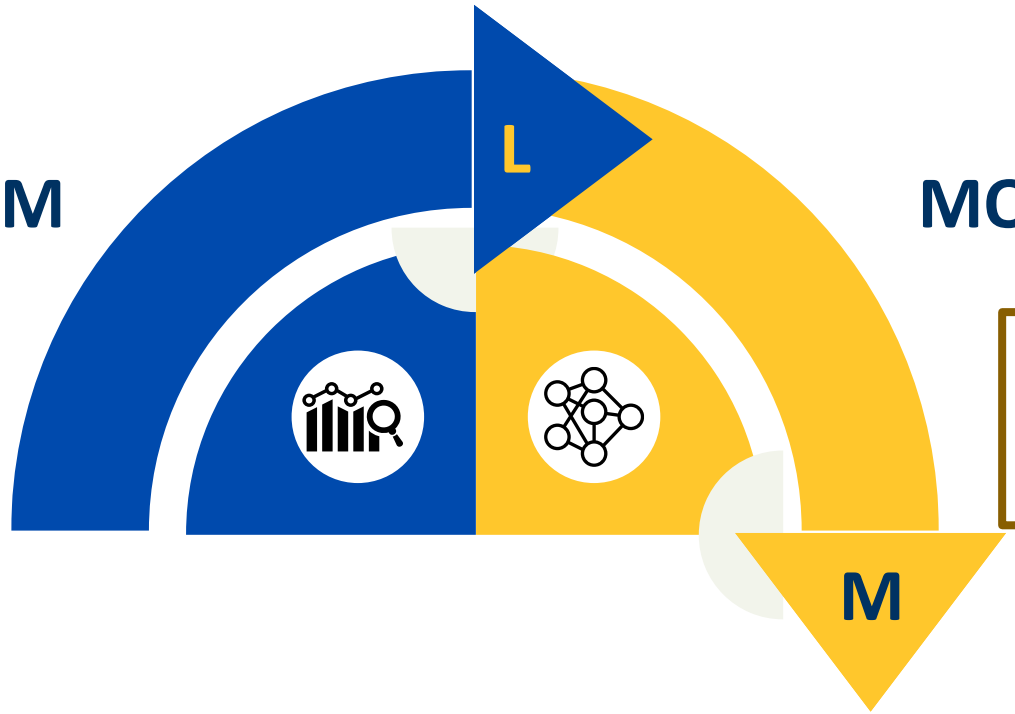


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Model Design

LEARNING PROBLEM

Supervised learning
of scalar target values



MODEL DESIGN

Linear Regression
with basis functions
$y(\vec{x}, \vec{w}) = x^T \vec{w}$

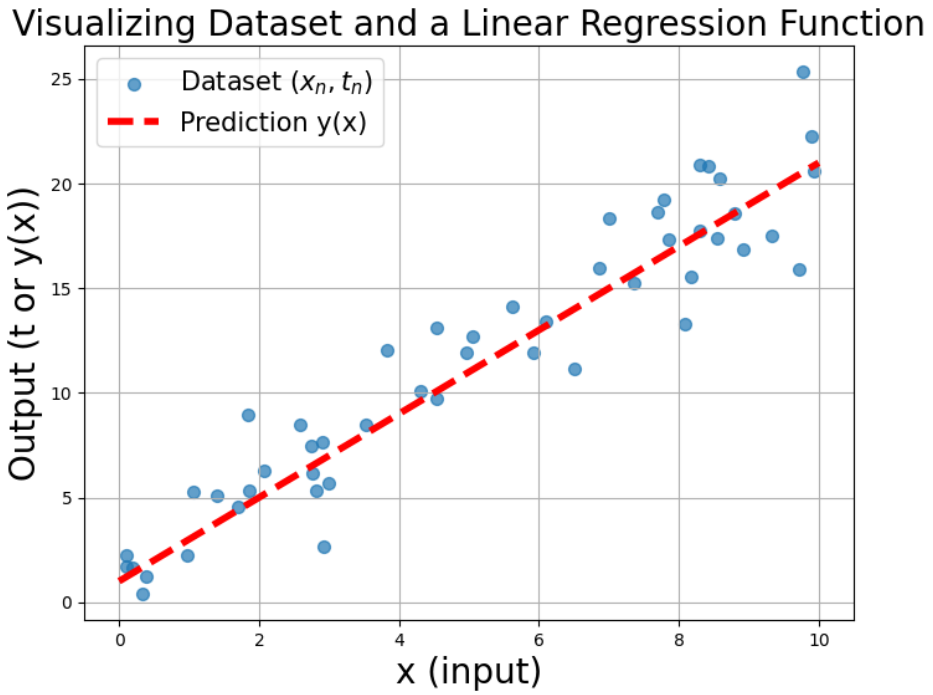
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Supervised Linear Regression

- Goal: Predict one or more continuous *target* variable t given the vector $\vec{x} \in \mathbb{R}^D$ of *input* variables.

$$\{(x_n)\}_{n=1}^N \xrightarrow{\vec{w}} \{(t_n)\}_{n=1}^N$$

- We formulate a function $y(\vec{x}, \vec{w})$ whose values for new input $\vec{x}$ constitute the predictions for the corresponding values of t.
- $\vec{w}$ is a vector of parameters that can be learned from the training data.



The Simplest Linear Regression Model

- The simplest model is of the following form:

$$y(\vec{\mathrm{x}}, \vec{w}) = w_0 + w_1 x_1 + \ \ldots + w_D x_D \quad \text{where } \vec{\mathrm{x}} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{R}^D.$$

Predicted output

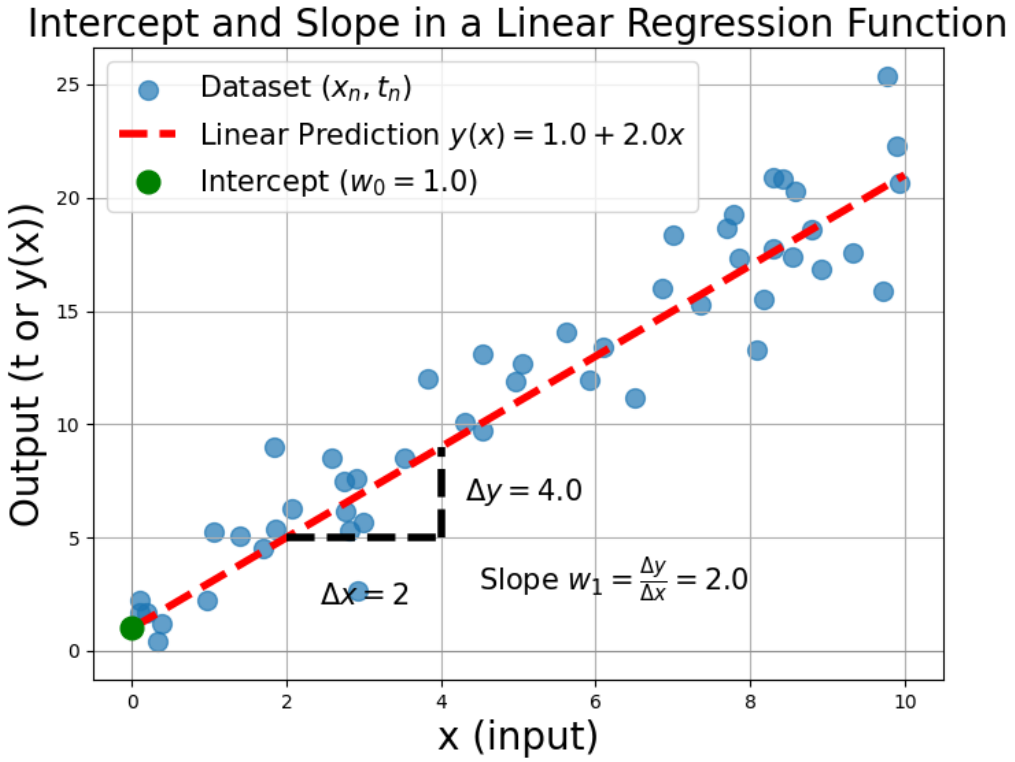
Linear combination of parameters $w_0, w_1, \ldots, w_D$, each scaled by x_d's.

- In the simplest case of $D = 1$, we have:

$$y(\vec{\mathrm{x}}, \vec{w}) = w_0 + w_1 x_1$$

Intercept (shift)

Slope (rate of change)



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The Simplest Linear Regression Model

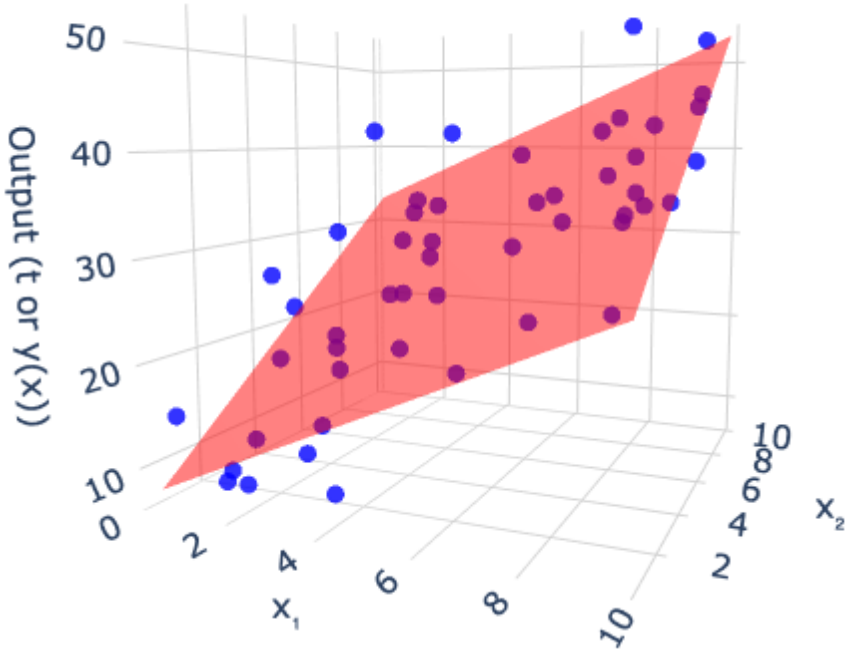
- The simplest model is of the following form:

$$y(\vec{x}, \vec{w}) = w_0 + w_1 x_1 + \ \ldots + w_D x_D \quad \text{where } \vec{x} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{R}^D.$$

Predicted output

Linear combination of parameters $w_0, w_1, \ldots, w_D$, each scaled by x_d's.

- In higher dimension, the predicted values form a hyperplane in $\mathbb{R}^{D+1}$.



Interactive link

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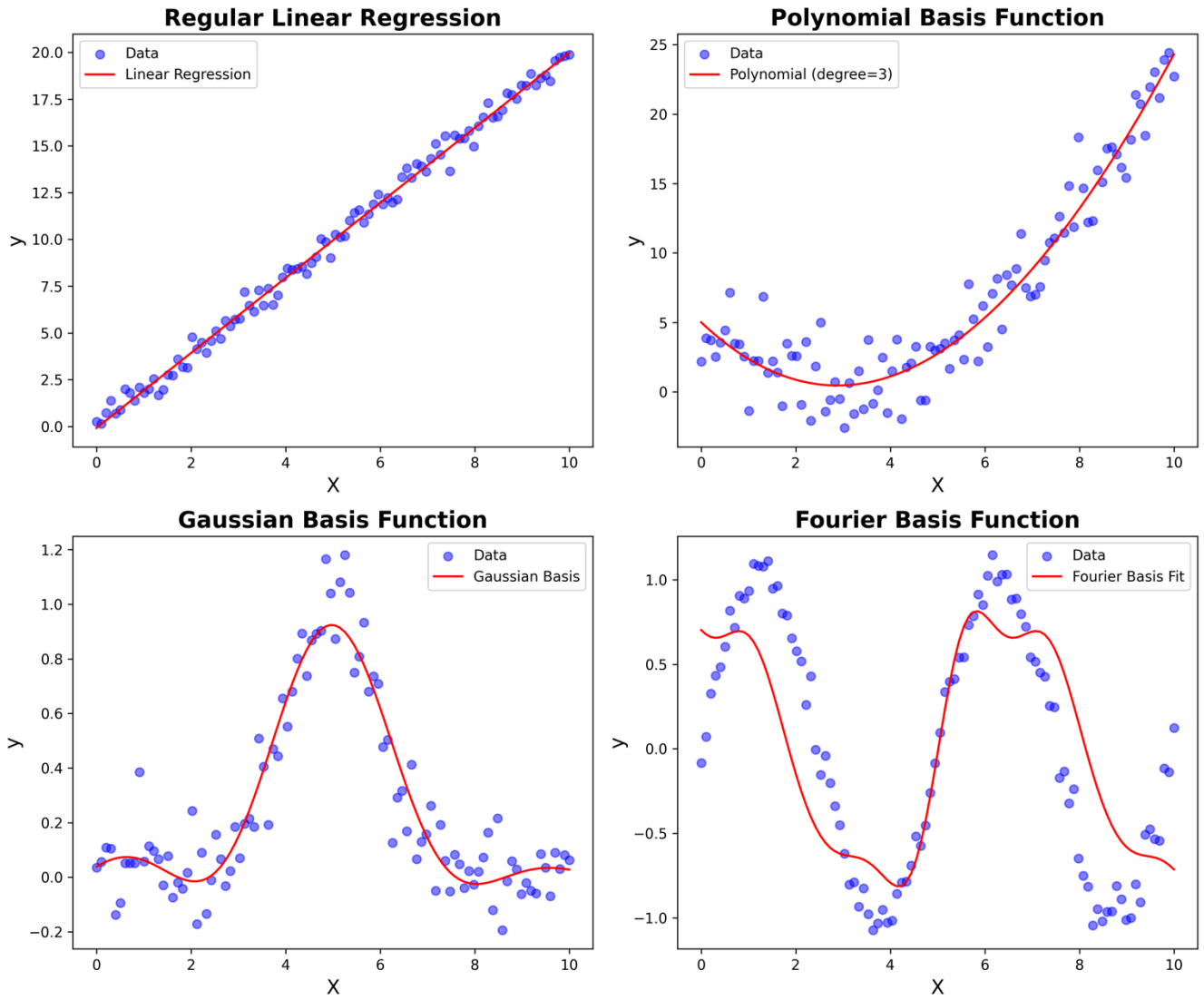
Basis Functions

- Linear Regression
- Basis Functions
- Vectorizing Calculations
- Error Function
- Error Function Minimization
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- Evaluation

Linear Functions From Slido

These are all linear models with different basis functions.

We will now see what basis functions are…



What Does It Mean To Be a Linear Model?

In what sense are the previous plots linearly modeled?

$$y(\vec{\mathrm{x}}, \vec{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathrm{x})$$

Are linear models linear in the

1. Features?
2. Parameters?

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What Does It Mean To Be a Linear Model?

In what sense are the previous plots linearly modeled?

$$y(\vec{\mathrm{x}}, \vec{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathrm{x})$$

Linear in the Parameters

Feature Functions

Are linear models linear in the

1. Features?
2. Parameters?

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Basis Functions

- We can extend the class of models by considering linear combinations of fixed nonlinear functions of the input variables, of the form:

$$y(\vec{\mathrm{x}}, \vec{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathrm{x})$$

where $\phi_j(\mathrm{x})$ are known as basis functions.

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More on Basis Function

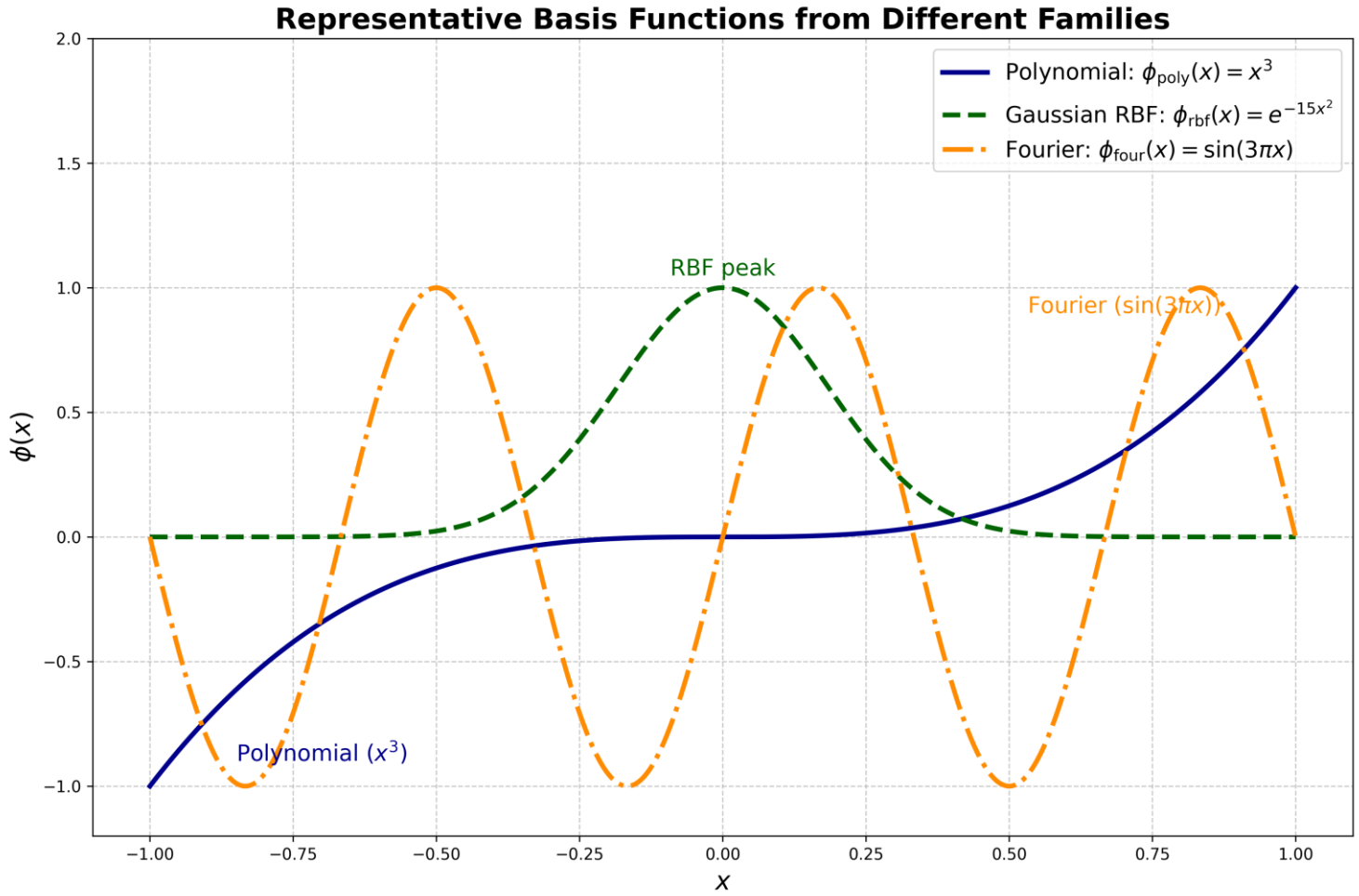
- **Polynomial** basis: $\phi_j(x) = x^j$
- **Radial Basis** Functions (RBF) such as **Gaussian**: $\phi_j(x) = e^{-\frac{(x-\mu_j)^2}{2\sigma^2}}$
- **Sigmoidal** basis: $\phi_j(x) = \sigma(\frac{x-\mu_j}{\sigma})$ where $\sigma(a) = \frac{1}{1+e^{-a}}$
  - $\tanh(a) = 2\sigma(2a) - 1$
- **Sinusoidal** basis: $\phi_j(x) = \sin x_j$ or $\phi_j(x) = cos\, x_j$

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Basis Functions as Features

Why did we care about basis functions before deep learning?

- Fixed feature extraction was the norm before deep learning: raw input x → hand-designed features $\phi_j(x)$.
- Goal: Pick expressive features so a simple linear model could succeed.
- Popular choices: polynomials, radial-basis functions, Fourier terms, kernels, etc.
- Heavy manual engineering.

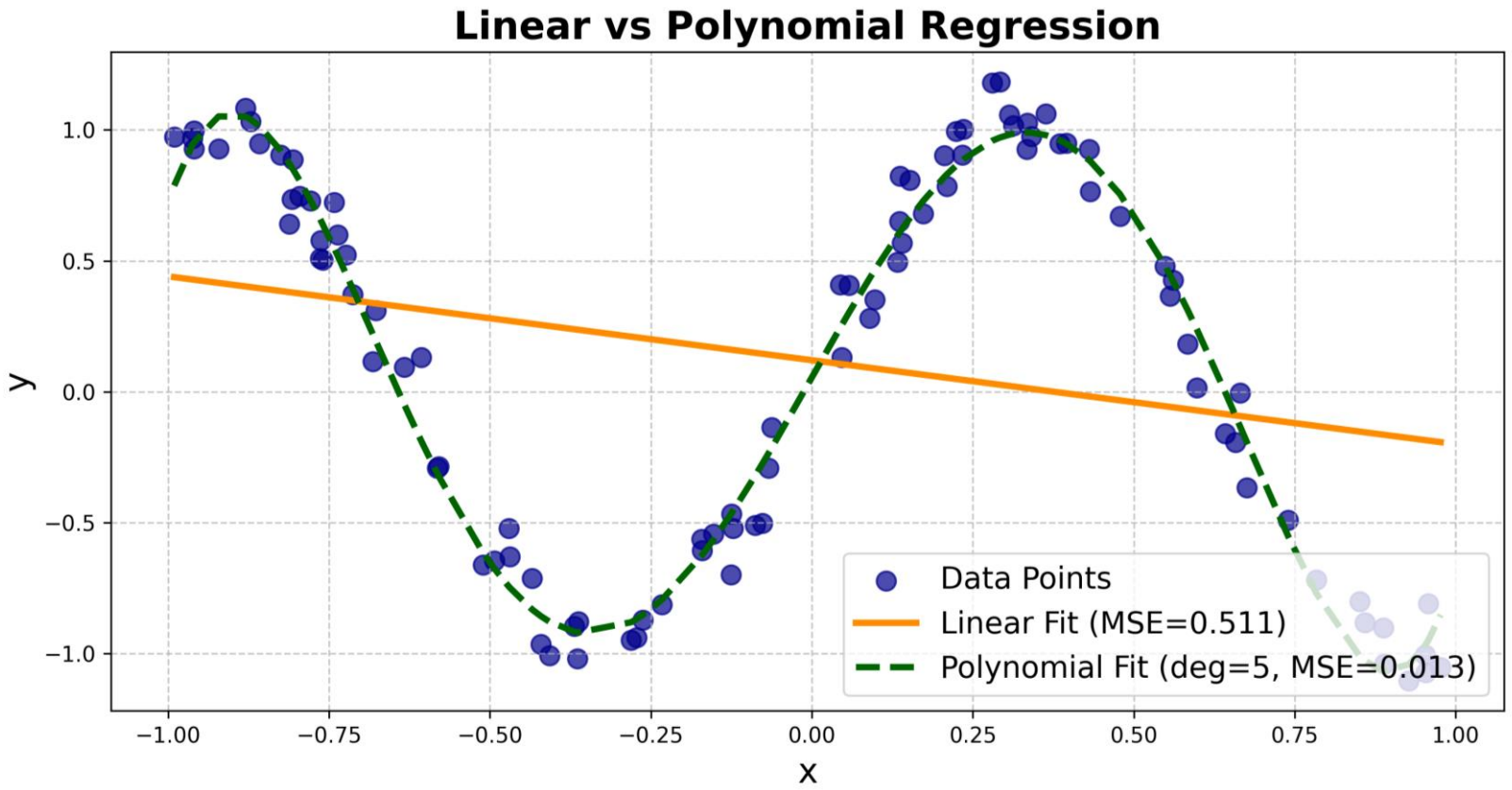


How Would Basis Functions Improve Predictions?

Using polynomial basis functions of degree 5, we redefined the linear regression equation as:

$$y(\vec{x}, \vec{w}) = \sum_{j=0}^{5} w_j x^j$$

Looking at the Mean Squared Error (MSE) between targets and predictions, the polynomial fit has a better performance.



$y(x) = 0.000 + 4.355x^1 - 0.203x^2 - 14.426x^3 + 0.164x^4 + 9.368x^5$

$y(x) = 0.120 - 0.320x$

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Vectorizing Calculations

- Linear Regression
- Basis Functions
- Vectorizing Calculations
- Error Function
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- Evaluation

Vectorizing Calculations

- Rewriting the equation for linear regression:

$$y(\vec{x}, \vec{w}) = w_0 + w_1 x_1 + \ \ldots + w_D x_D = w_0 + \sum_{d=1}^{D} w_d x_d$$

- If we assume that there is an $x_0 = 1$, then the two terms above can be combined into one summation:

$$y(\vec{x}, \vec{w}) = w_0 + w_1 x_1 + \ \ldots + w_D x_D = \sum_{d=0}^{D} w_d x_d$$

where $\vec{x} = (x_0, x_1, x_2, \ldots, x_D)^T$ and $\vec{w} = (w_0, w_1, w_2, \ldots, w_D)^T$

Augmented input with x_0 $\quad$ $x_0 = 1$

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Vectorizing Calculations

$$y(\vec{x}, \vec{w}) = w_0 + w_1 x_1 + \ \ldots + w_D x_D = \sum_{d=0}^{D} w_d x_d$$

where $\vec{x} = (x_0, x_1, x_2, \ldots, x_D)^T$ and $\vec{w} = (w_0, w_1, w_2, \ldots, w_D)^T$

$$y(\vec{x}, \vec{w}) = \vec{w} \cdot \vec{x}$$

$$= \vec{w}^T \vec{x} = \vec{x}^T \vec{w}$$

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Matrix Notation

- Assuming we have $n = 1, \dots, N$ samples in our dataset, we can write:

$$\begin{cases} y(\overrightarrow{x_1}, \vec{w}) = w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_D x_{1D} \\ y(\overrightarrow{x_2}, \vec{w}) = w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_D x_{2D} \\ \dots \\ y(\overrightarrow{x_N}, \vec{w}) = w_0 + w_1 x_{N1} + w_2 x_{N2} + \dots + w_D x_{ND} \end{cases}$$

$$\begin{cases} y(\overrightarrow{x_1}, \vec{w}) = \overrightarrow{\mathbf{x}_1^T}\, \vec{w} & \text{where } \overrightarrow{\mathbf{x}_1^T} = [1, x_{11}, x_{12}, \dots, x_{1D}] \\ y(\overrightarrow{x_2}, \vec{w}) = \overrightarrow{\mathbf{x}_2^T}\, \vec{w} & \text{where } \overrightarrow{\mathbf{x}_2^T} = [1, x_{21}, x_{22}, \dots, x_{2D}] \\ \dots \\ y(\overrightarrow{x_N}, \vec{w}) = \overrightarrow{\mathbf{x}_N^T}\, \vec{w} & \text{where } \overrightarrow{\mathbf{x}_N^T} = [1, x_{N1}, x_{N2}, \dots, x_{ND}] \end{cases}$$

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Matrix Notation

$$\begin{cases} y(\overrightarrow{x_1}, \vec{w}) = \overrightarrow{\mathbf{x}_1^T}\,\vec{w} & \text{where } \overrightarrow{\mathbf{x}_1^T} = [1, x_{11}, x_{12}, \dots, x_{1D}] \\ y(\overrightarrow{x_2}, \vec{w}) = \overrightarrow{\mathbf{x}_2^T}\,\vec{w} & \text{where } \overrightarrow{\mathbf{x}_2^T} = [1, x_{21}, x_{22}, \dots, x_{2D}] \\ \dots & \\ y(\overrightarrow{x_N}, \vec{w}) = \overrightarrow{\mathbf{x}_N^T}\,\vec{w} & \text{where } \overrightarrow{\mathbf{x}_N^T} = [1, x_{N1}, x_{N2}, \dots, x_{ND}] \end{cases}$$

$$\begin{cases} y(\overrightarrow{x_1}, \vec{w}) = [\,1, x_{11}, x_{12}, \dots, x_{1D}\,]\,\vec{w} \\ y(\overrightarrow{x_2}, \vec{w}) = [\,1, x_{21}, x_{22}, \dots, x_{2D}\,]\,\vec{w} \\ \dots \\ y(\overrightarrow{x_N}, \vec{w}) = [1, x_{N1}, x_{N2}, \dots, x_{ND}]\,\vec{w} \end{cases}$$

$$\mathbb{X} = \{\overrightarrow{x_1}, \overrightarrow{x_2}, \dots, \overrightarrow{x_N}\} \in \mathbb{R}^{N\times(D+1)}$$

Design matrix

$$\mathbb{Y} = \{y(\overrightarrow{x_1}, \vec{w}), y(\overrightarrow{x_2}, \vec{w}), \dots, y(\overrightarrow{x_N}, \vec{w})\} \in \mathbb{R}^N$$

$$\mathbb{Y}(\mathbb{X}, \vec{w}) = \mathbb{X}\,\vec{w}$$

$\in \mathbb{R}^{D+1}$

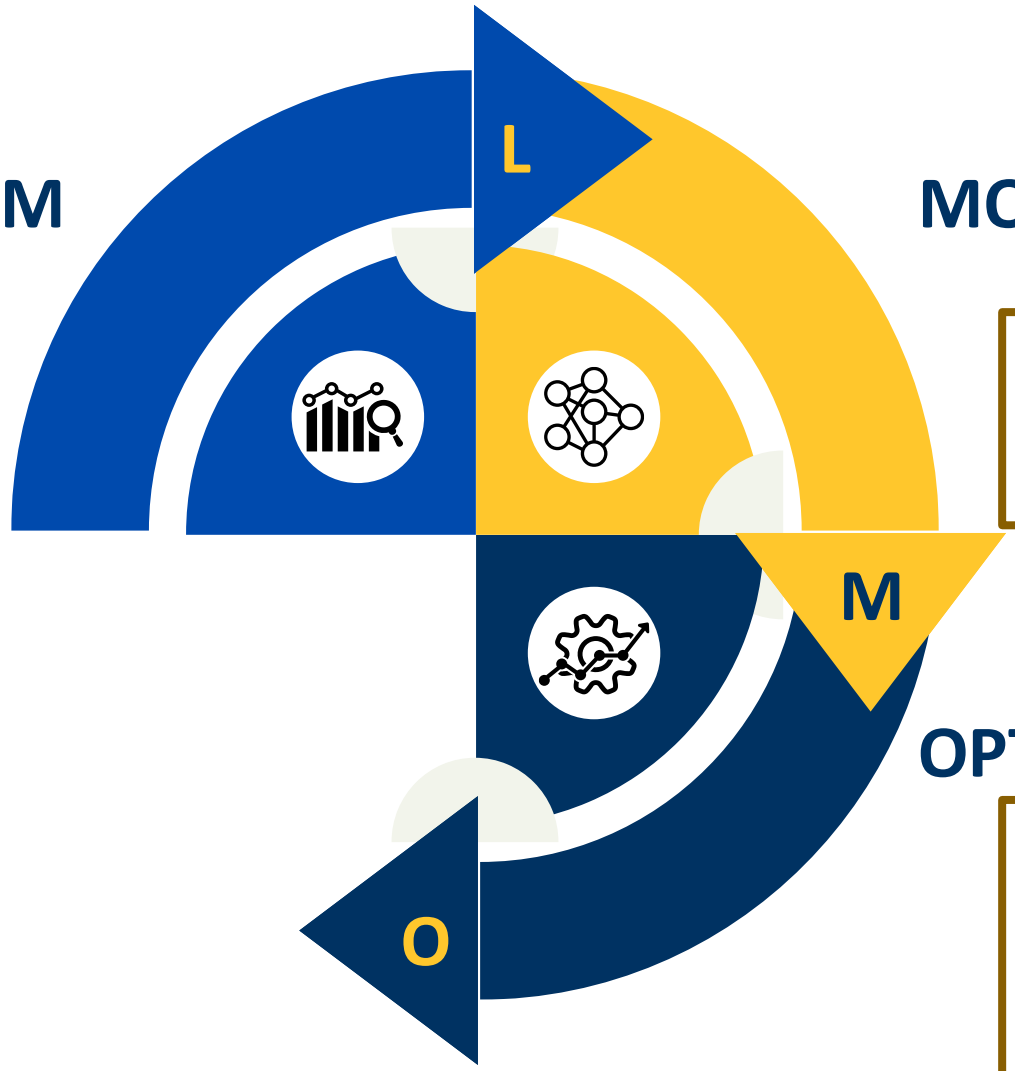
$\in \mathbb{R}^{N\times(D+1)}$

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Error Function

- Linear Regression
- Basis Functions
- Vectorizing Calculations
- Error Function
- Error Function Minimization
- Geometric Interpretation
- Evaluation

Optimization



LEARNING PROBLEM

Supervised learning of scalar target values

MODEL DESIGN

Linear Regression with basis functions

$$y(\vec{x}, \vec{w}) = x^T \vec{w}$$

OPTIMIZATION

Derive direct solution

$$E(w) = \frac{1}{2} \sum_{n=1}^{N} \left(t_n - y(x_n, w) \right)^2$$

$$w^* = \underset{w}{\operatorname{argmin}}\, E(w)$$

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Error Function

- How to calculate the weights $\vec{w}$?
- We do this by minimizing an ***error function*** that measures the **misfit** between the function $\mathbb{Y}(\mathbb{X}, \vec{w})$ and **targets**, for any given value of $\vec{w}$, and the training data $\mathbb{X}$.
- One simple error function: **Sum of the squares of the differences between targets and the predictions.**

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - y(\mathrm{x}_n, w)\right)^2$$

- Non-negative quantity.
- Only 0 if all predictions are equal to targets.

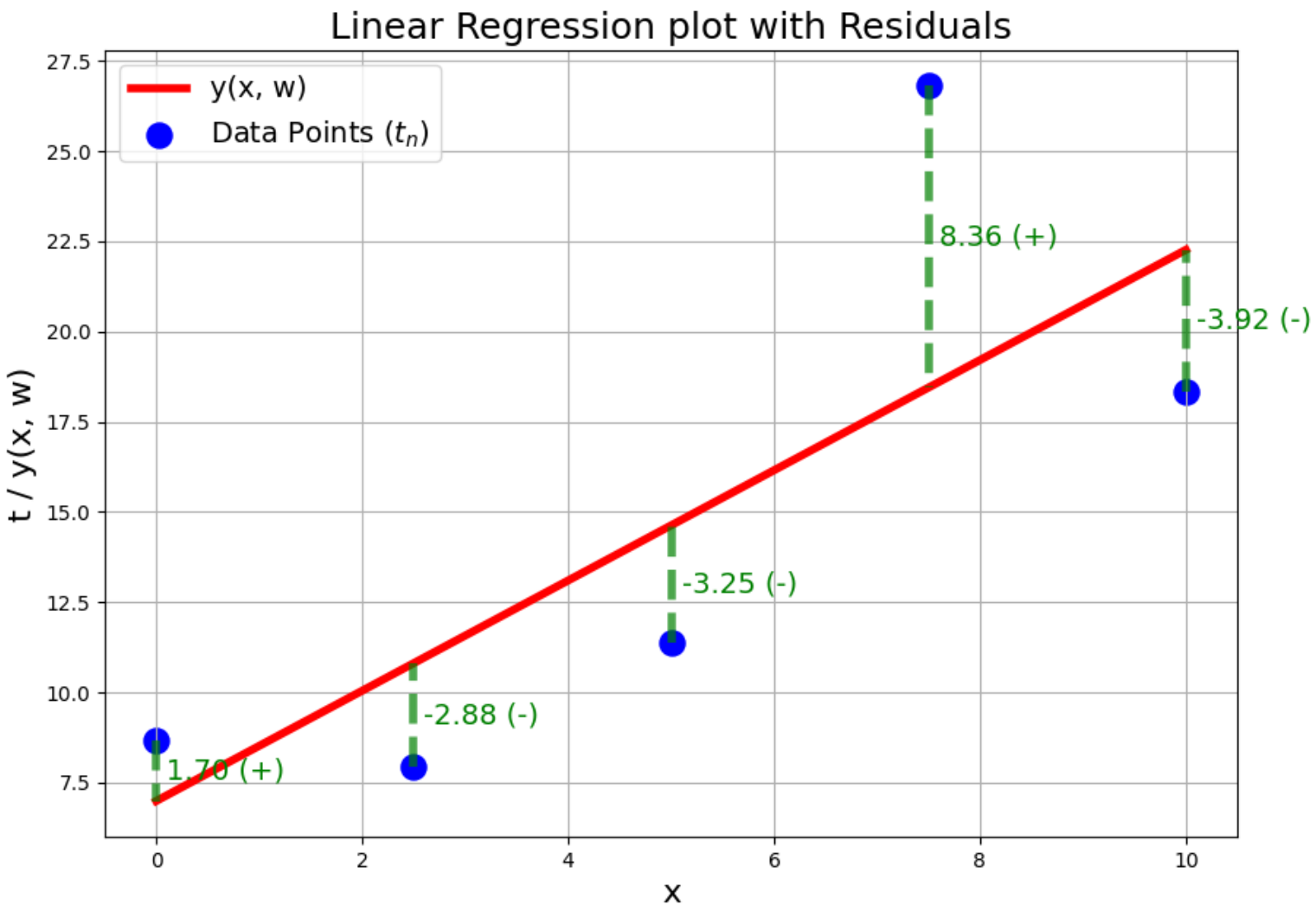
The factor of $\frac{1}{2}$ is included for later convenience.

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Error Function Visualization

Residuals are defined as $e_n = (t_n - y(x_n, w))$ and they can be + or -.

The error function $E(w)$ combines the squares of the residuals to measure the goodness of the fit.



Error Function Minimization

- Gaussian Mixture Model
- Linear Regression
- Basis Functions
- Vectorizing Calculations
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Error Function Minimization

- $E(w)$ is a quadratic function.

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2$$

Therefore, $\frac{\partial E(w)}{\partial w}$ is a linear function of w and hence solving $\frac{\partial E(w)}{\partial w} = 0$ has one and only one solution. Let's now derive this solution:

$$= \frac{1}{2}\sum_{n=1}^{N}\left(t_n - w_0 - \sum_{j=1}^{D} w_j x_{nj}\right)^2$$

Finding the optimum solution

$$\frac{\partial E(w)}{\partial w} = 0$$

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Error Function Minimization

$$\frac{\partial E(w)}{\partial w_0} = 0 \longrightarrow \frac{\partial\left(\frac{1}{2}\sum_{n=1}^{N}\left(\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)\right)^2\right.}{\partial w_0} = 0$$

Sum rule

$$\partial \sum f = \sum \partial f$$

$$= \frac{\frac{1}{2}\sum_{n=1}^{N} \partial\left(\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)^2\right)}{\partial w_0}$$

Chain rule

$$\partial f^2 = 2f\partial f$$

$$= \frac{1}{2}\sum_{n=1}^{N} 2\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)\left(\frac{\partial\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)}{\partial w_0}\right)$$

Separating the terms

$$= \sum_{n=1}^{N}\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)(-1) = -\left(\sum_{n=1}^{N} t_n - nw_0 - \sum_{n=1}^{N}\left(\sum_{d=1}^{D} w_d x_{nd}\right)\right)$$

$$\frac{\partial E(w)}{\partial w_0} = 0 \longrightarrow w_0 = \frac{1}{n}\left(\sum_{n=1}^{N} t_n - \sum_{n=1}^{N}\left(\sum_{d=1}^{D} w_d x_{nd}\right)\right)$$

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Error Function Minimization

$$\frac{\partial E(w)}{\partial w_k} = 0 \longrightarrow \frac{\partial\left(\frac{1}{2}\sum_{n=1}^{N}\left(\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)\right)^2\right)}{\partial w_k} = 0$$

Sum rule

$$\partial \sum f = \sum \partial f$$

$$= \frac{\frac{1}{2}\sum_{n=1}^{N} \partial\left(\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)^2\right)}{\partial w_k}$$

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Reordering

$$= \sum_{n=1}^{N}\left(t_n - w_0 - \sum_{d=1}^{D} w_d x_{nd}\right)(-x_{nk})$$

$$\frac{\partial E(w)}{\partial w_k} = -\sum_{n=1}^{N} x_{nk}\left(t_n - w_0 - \sum_{d=1}^{D} w_d\, x_{nd}\right)$$

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Error Function Minimization

$$\frac{\partial E(w)}{\partial w_k} = -\sum_{n=1}^{N} x_{nk}\left(t_n - w_0 - \sum_{d=1}^{D} w_d\, x_{nd}\right)$$

$$\sum_{n=1}^{N} x_{nk} t_n = \sum_{n=1}^{N} x_{nk}\left(w_0 + \sum_{d=1}^{D} w_d x_{nd}\right) \longrightarrow \underbrace{\sum_{n=1}^{N} x_{nk} t_n}_{\mathbb{X}^T T} = \underbrace{w_0 \sum_{n=1}^{N} x_{nk} + \sum_{d=1}^{D} w_d \sum_{n=1}^{N} x_{nk}\, x_{nd}}_{\mathbb{X}^T \mathbb{X}\, w}$$

Normal equations for the least squares problem

$$\mathbb{X}^T T = \mathbb{X}^T \mathbb{X}\, w$$

Multiplying both sides by $(\mathbb{X}^T \mathbb{X})^{-1}$

$$(\mathbb{X}^T \mathbb{X})^{-1}\, \mathbb{X}^T T = w$$

Takeaway

$$w^* = (\mathbb{X}^T \mathbb{X})^{-1}\, \mathbb{X}^T T$$

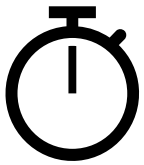
Assuming $\mathbb{X}^T \mathbb{X}$ is invertible

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Solving Normal Equation Runtime

$$\mathbb{X} = \{\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_N}\} = \underbrace{\begin{bmatrix} 1 & x_{11} & \ldots & x_{1D} \\ 1 & x_{21} & \ldots & x_{2D} \\ \ldots & \ldots & \ldots & \ldots \\ 1 & x_{N1} & \ldots & x_{ND} \end{bmatrix}}_{D+1} \Big\} N \quad \in \mathbb{R}^{N \times (D+1)}$$

$$\mathbb{X}^T\mathbb{X} \in \mathbb{R}^{(D+1)\times(D+1)}$$



Time complexity

Forming $\mathbb{X}^T\mathbb{X}$: $\mathcal{O}(N^2D^2)$

Forming $\mathbb{X}^T T$: $\mathcal{O}(N^2D)$

Forming $(\mathbb{X}^T\mathbb{X})^{-1}$: $\mathcal{O}(D^3)$

Calculating $(\mathbb{X}^T\mathbb{X})^{-1}\,\mathbb{X}^T T$: $\mathcal{O}(D^3)$

$$\rightarrow \mathcal{O}(N^2D^2 + D^3)$$

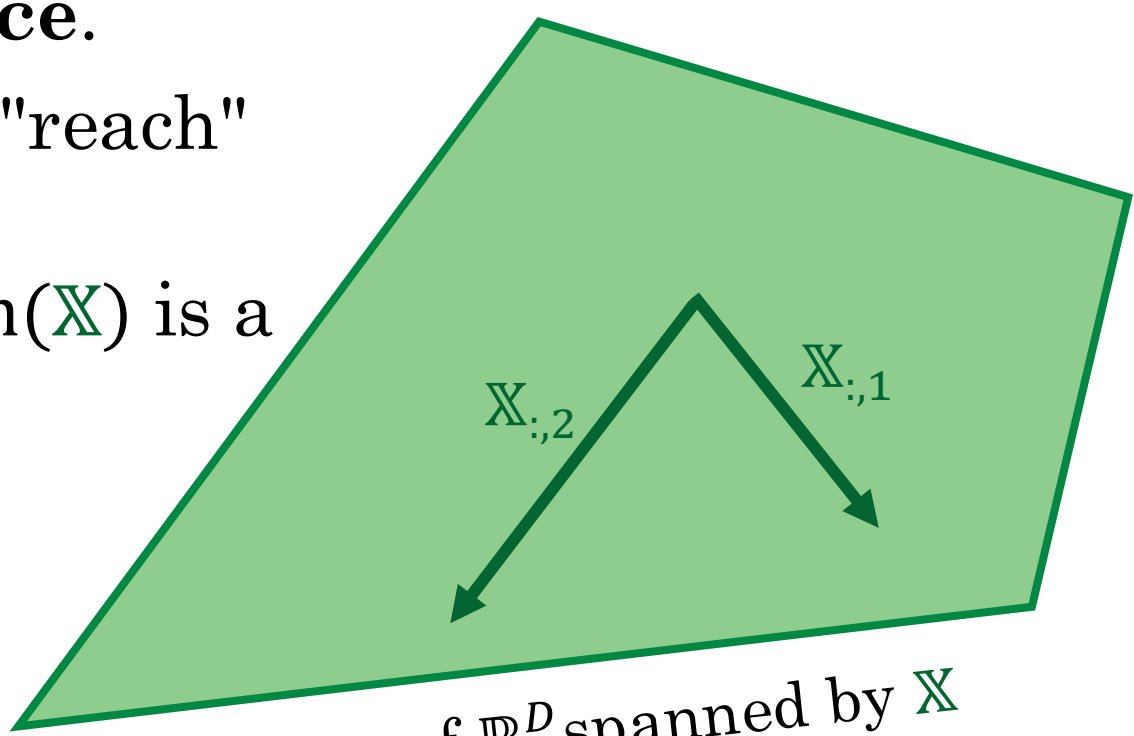
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Geometric Interpretation

- Linear Regression
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[Linear Algebra] Span

- The set of all possible linear combinations of the columns of $\mathbb{X}$ is called the **span** of $\mathbb{X}$ (denoted span($\mathbb{X}$)), also called the **column space**.
- Intuitively, this is all vectors you can "reach" using the columns of $\mathbb{X}$.
- If each column of $\mathbb{X}$ has length D, span($\mathbb{X}$) is a subspace of $\mathbb{R}^D$.



Subspace of $\mathbb{R}^D$ spanned by $\mathbb{X}$

[Linear Algebra] Matrix-Vector Multiplication

- **Approach 1**: So far, we've thought of our model as horizontally stacked predictions per datapoint:

$$\begin{bmatrix} | \\ \mathbb{Y} \\ | \end{bmatrix} = \begin{bmatrix} — & \mathbf{x}_1^T & — \\ — & \mathbf{x}_2^T & — \\ & \vdots & \\ — & \mathbf{x}_N^T & — \end{bmatrix} \begin{bmatrix} \\ w \\ \\ \end{bmatrix}$$

Matrix sizes
$N \times 1 = (N \times (D+1))\ ((D+1) \times 1)$

- **Approach 2**: However, it is helpful sometimes to think of matrix-vector multiplication as performed by columns. We can also think of $\mathbb{Y}$ as a **linear combination of feature vectors**, scaled by **parameters.**

$$\begin{bmatrix} | \\ \mathbb{Y} \\ | \end{bmatrix} = \begin{bmatrix} | & & | \\ \mathbb{X}_{:,0} & \cdots & \mathbb{X}_{:,D} \\ | & & | \end{bmatrix} \begin{bmatrix} \\ w \\ \\ \end{bmatrix} = w_0 \begin{bmatrix} | \\ \mathbb{X}_{:,0} \\ | \end{bmatrix} + w_1 \begin{bmatrix} | \\ \mathbb{X}_{:,1} \\ | \end{bmatrix} + \cdots + w_D \begin{bmatrix} | \\ \mathbb{X}_{:,D} \\ | \end{bmatrix}$$

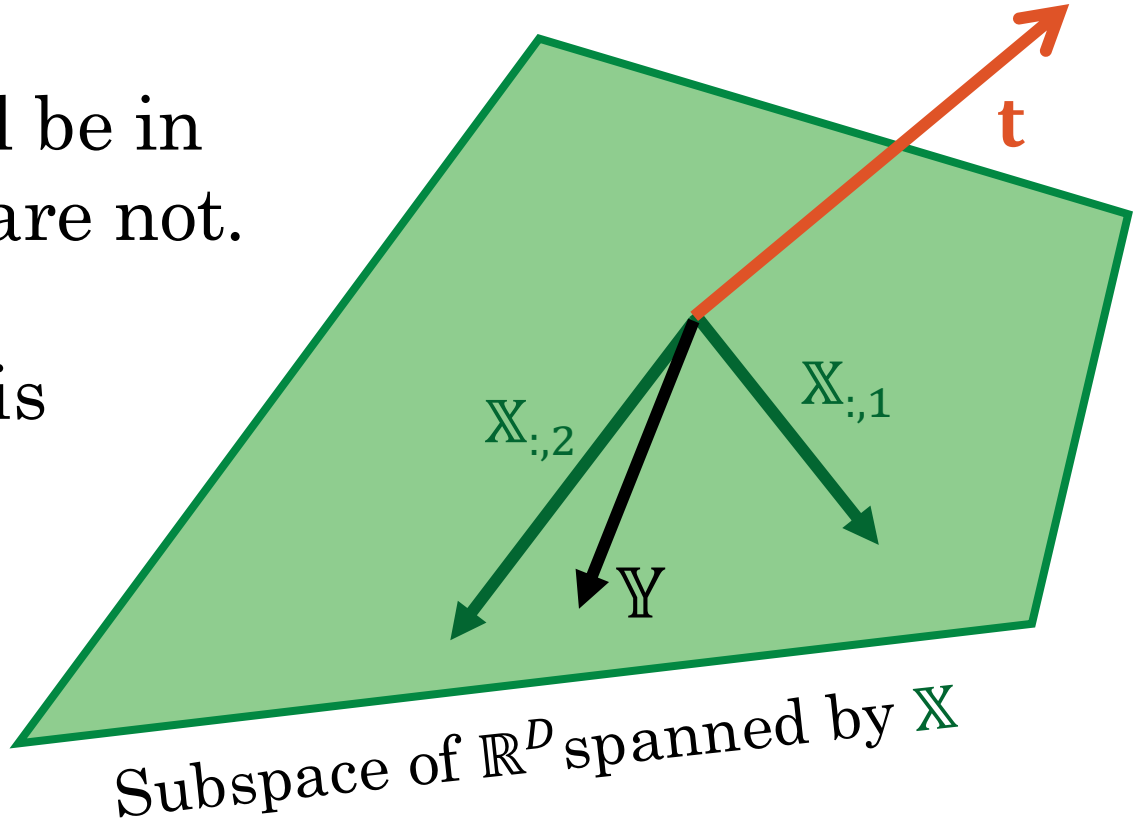
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Prediction Is a Linear Combination of Columns

Our prediction of $\mathbb{Y}(\mathbb{X}, \vec{w}) = \mathbb{X}\,\vec{w}$ is a linear combination of columns of $\mathbb{X}$.

Interpret: Our linear prediction $\mathbb{Y}$ will be in span($\mathbb{X}$), even if ground-truth values **t** are not.

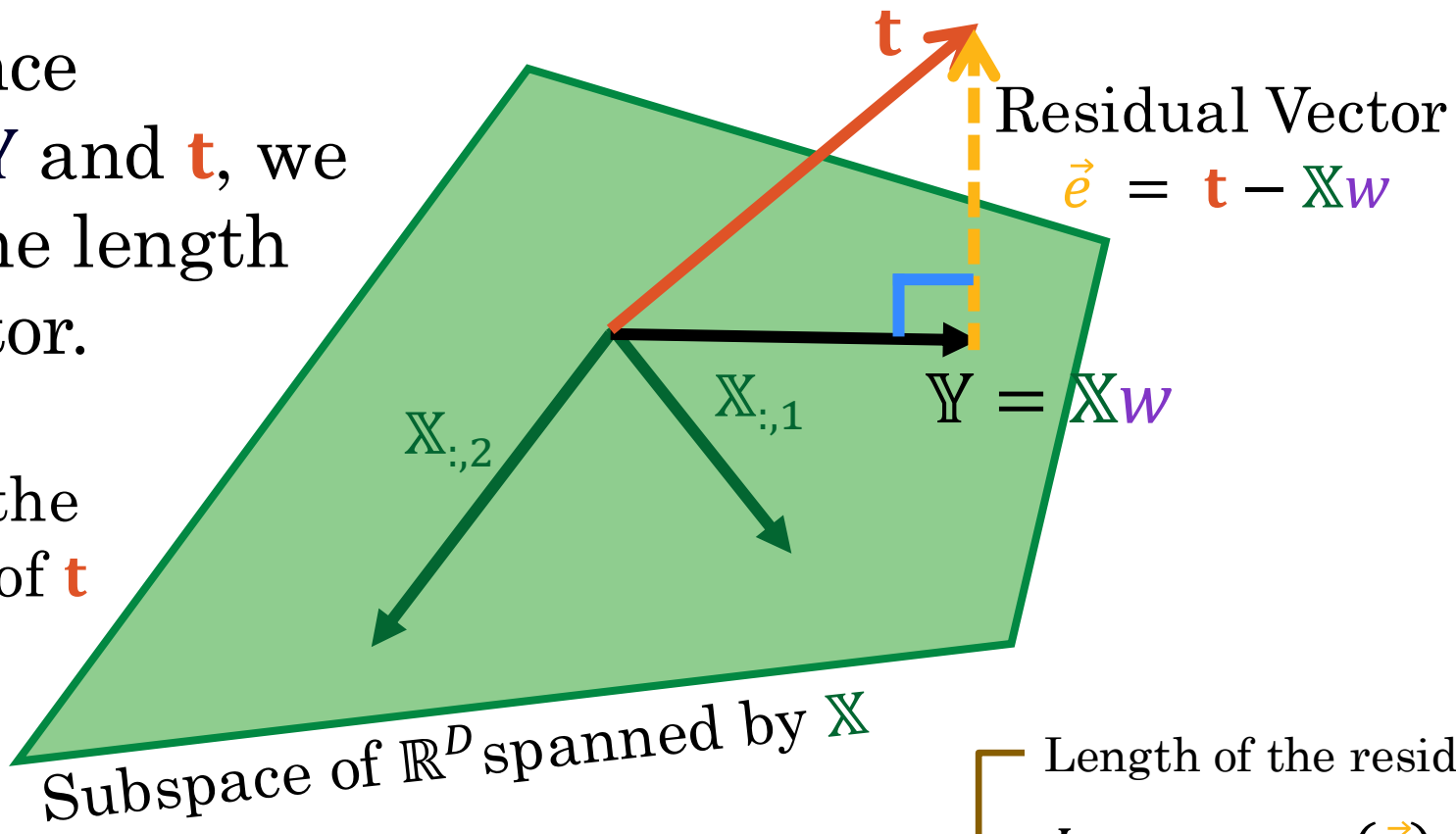
Goal: Find vector of $\mathbb{Y}$ in span($\mathbb{X}$) that is closest to **t**.



Finding Optimum Predictions ($\mathbb{Y}$)

To minimize distance between vector of $\mathbb{Y}$ and **t**, we should minimize the length of the residual vector.

$\vec{e}$ is minimized if it is the orthogonal projection of **t** on the span($\mathbb{X}$).

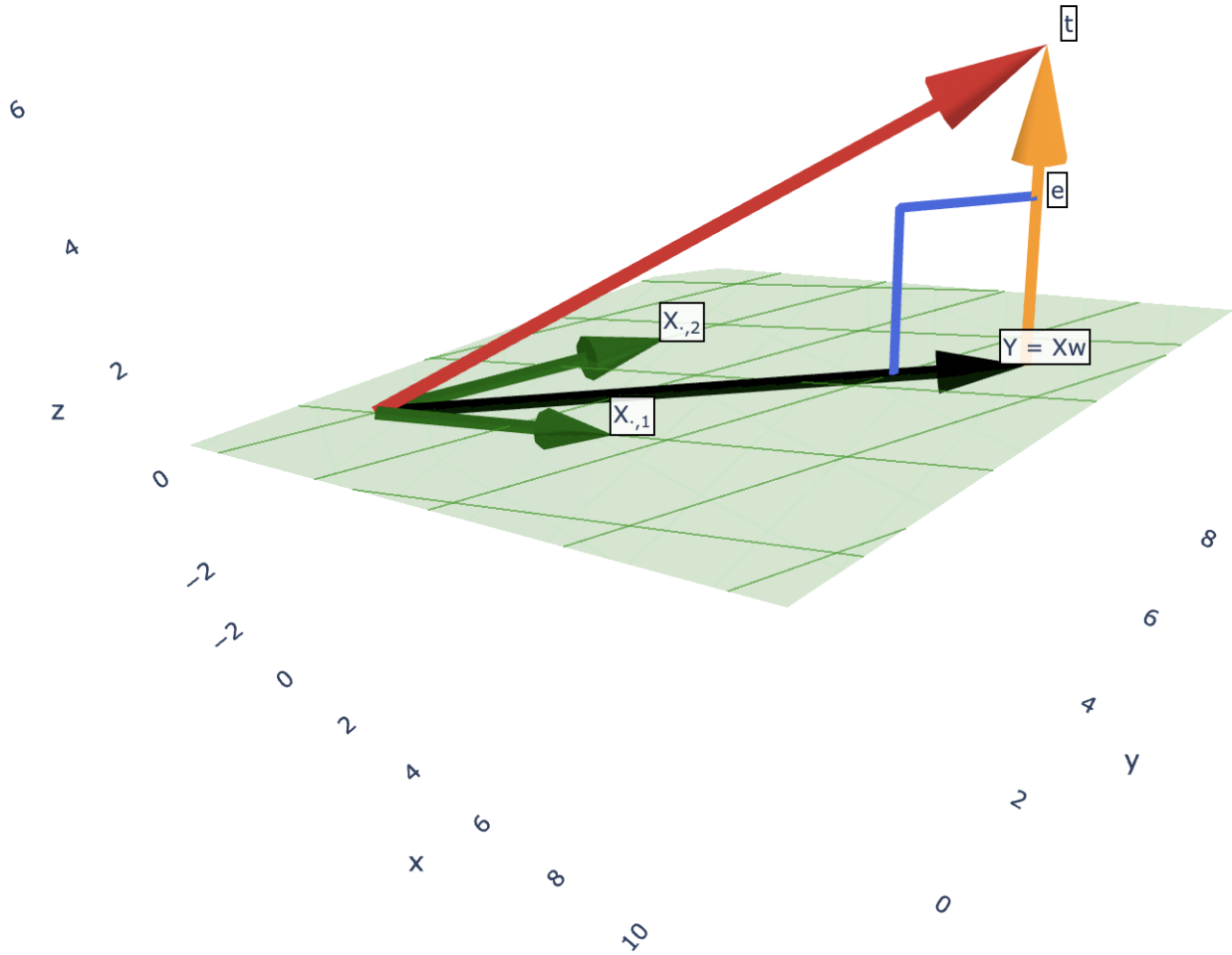


Length of the residual vector

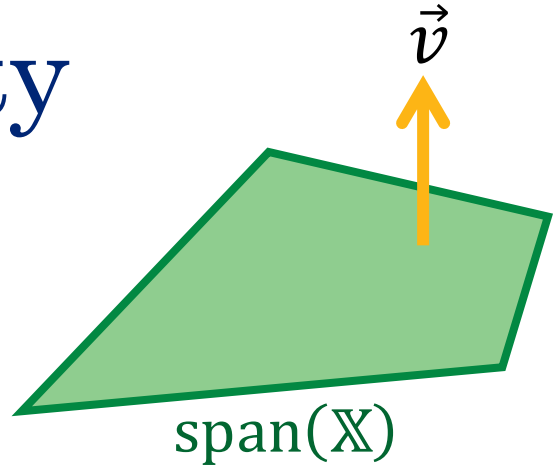
$$L_2 - norm\,(\vec{e}) = \|\vec{e}\|_2 = \sqrt{\sum_{d=0}^{D} e_d^2}$$

Geometry of Least Squares in Plotly

Interactive link



[Linear Algebra] Orthogonality



- A vector $\vec{v}$ is orthogonal to the span($\mathbb{X}$) if and only if it is orthogonal to all columns of $\mathbb{X}$.

Assuming $\vec{v} \in \mathbb{R}^N$ and $\mathbb{X} = \begin{bmatrix} | & & | \\ \mathbb{X}_{:,0} & \cdots & \mathbb{X}_{:,D} \\ | & & | \end{bmatrix} \in \mathbb{R}^{N \times D}$

$\vec{v}$ orthogonal to each column means:

$$\begin{aligned} \mathbb{X}_{:,0}^T \vec{v} &= 0 \\ \mathbb{X}_{:,1}^T \vec{v} &= 0 \\ &\vdots \\ \mathbb{X}_{:,D}^T \vec{v} &= 0 \end{aligned} \quad \Rightarrow \quad \begin{bmatrix} \mathbb{X}_{:,0}^T \vec{v} \\ \mathbb{X}_{:,1}^T \vec{v} \\ \vdots \\ \mathbb{X}_{:,D}^T \vec{v} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \Rightarrow \quad \mathbb{X}^T \vec{v} = 0$$

We will use this shortly

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Going Back to Our Error Function

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - y(\mathrm{x}_n, w)\right)^2 = \frac{1}{2}(\|\vec{e}\|_2)^2$$

Equivalently, vector $\vec{e}$ is orthogonal to the span($\mathbb{X}$), meaning that:

$\mathbb{X}^T\vec{e} = 0$ Adding the definition of residual

$\mathbb{X}^T\left(\mathbf{t} - \mathbb{Y}(\mathbb{X}, w^*)\right) = 0$ Adding formula for $\mathbb{Y}(\mathbb{X}, w^*) = \mathbb{X}w^*$

$\mathbb{X}^T\mathbf{t} - \mathbb{X}^T\mathbb{X}w^* = 0$ Moving terms

$\mathbb{X}^T\mathbf{t} = \mathbb{X}^T\mathbb{X}w^*$

Normal Equation

$w^* = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbf{t}$

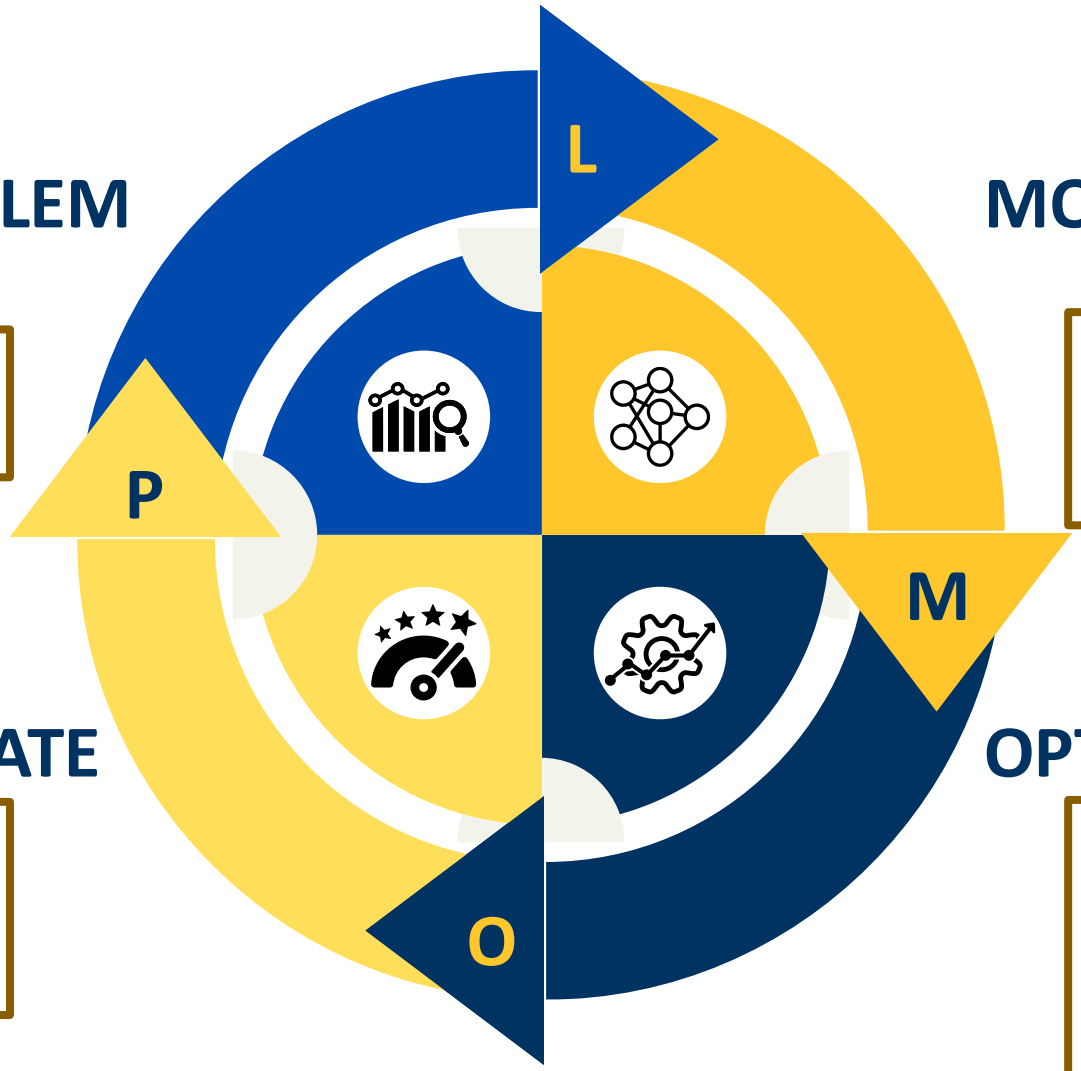
If $\mathbb{X}^T\mathbb{X}$ is invertible

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Evaluation

- Linear Regression
- Basis Functions
- Vectorizing Calculations
- Error Function
- Error Function Minimization
- Geometric Interpretation
- Evaluation

Predict and Evaluate



LEARNING PROBLEM

Supervised learning
of scalar target values

MODEL DESIGN

Linear Regression
with basis functions

$$y(\vec{x}, \vec{w}) = x^T \vec{w}$$

PREDICT & EVALUATE

Evaluate Predictions

$$\hat{y} = x_{test}^T w^*$$

and test error

OPTIMIZATION

Derive direct solution

$$E(w) = \frac{1}{2} \sum_{n=1}^{N} \left(t_n - y(x_n, w) \right)^2$$

$$w^* = \underset{w}{\operatorname{argmin}}\, E(w)$$

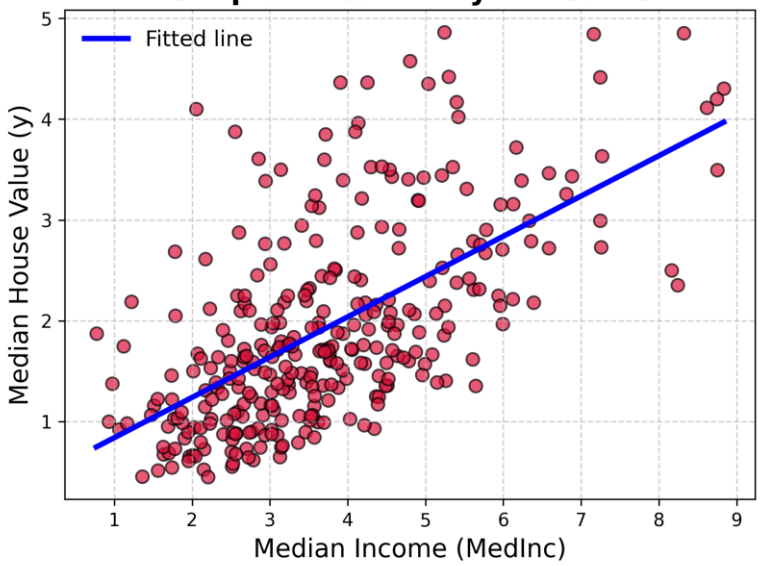
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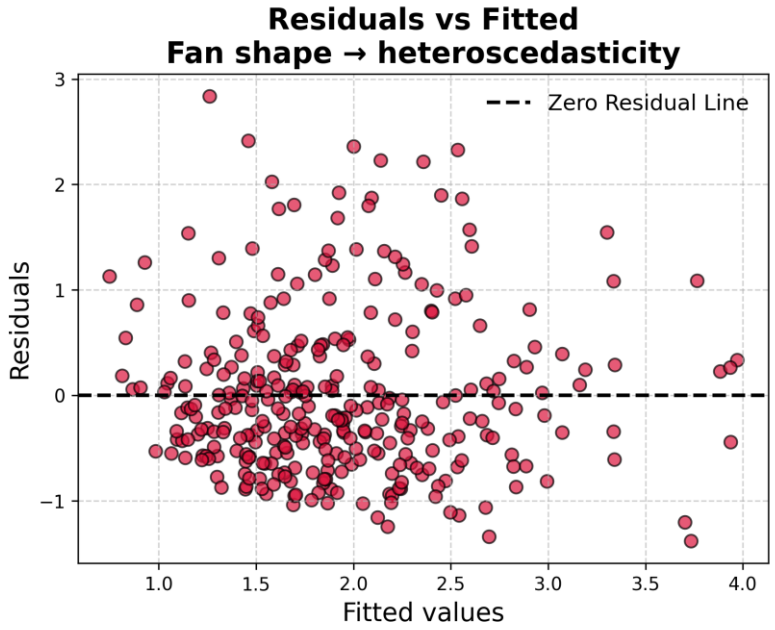
Evaluation - Visualization

Residual plot shows the trend of the residuals $e_n = (t_n - y(x_n, w))$ with respect to predictions $y(x_n, w)$.

What a good residual plot looks like:

- Points scattered randomly around 0 (no pattern).
- Roughly constant vertical spread (homoscedasticity).
- No obvious trend with fitted values or with predictors.





Evaluation - Metrics

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Evaluation - Metrics

Mean Squared Error (MSE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(\mathrm{x}_n, w)\right)^2\right)$$

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Evaluation - Metrics

Mean Squared Error (MSE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)$$

Root Mean Squared Error (RMSE)

$$\sqrt{\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)}$$

Moves the metric back to the original unit of the data compared to MSE

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Evaluation - Metrics

Mean Squared Error (MSE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)$$

Root Mean Squared Error (RMSE)

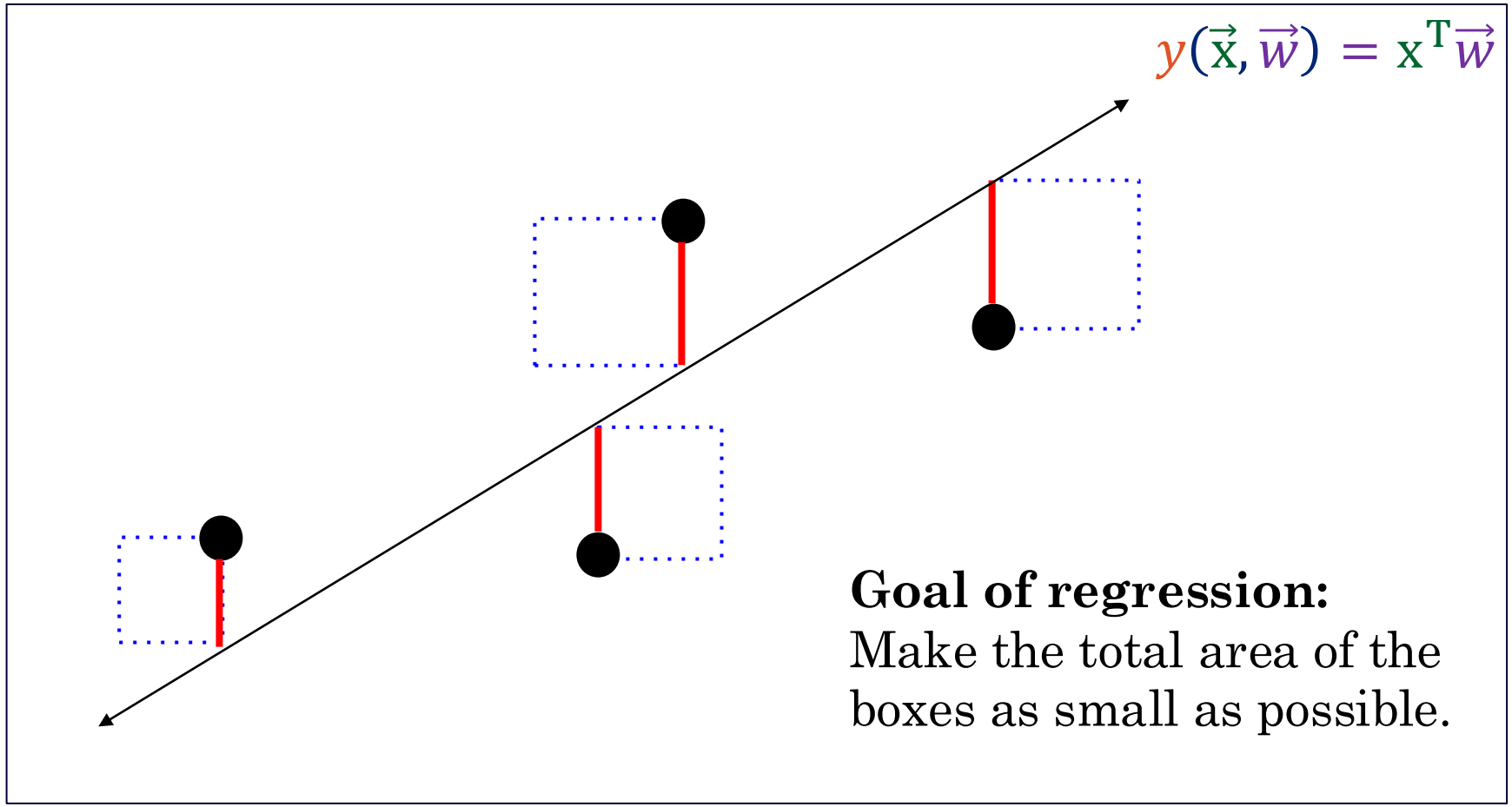
$$\sqrt{\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)}$$

R-Squared (R^2) Score

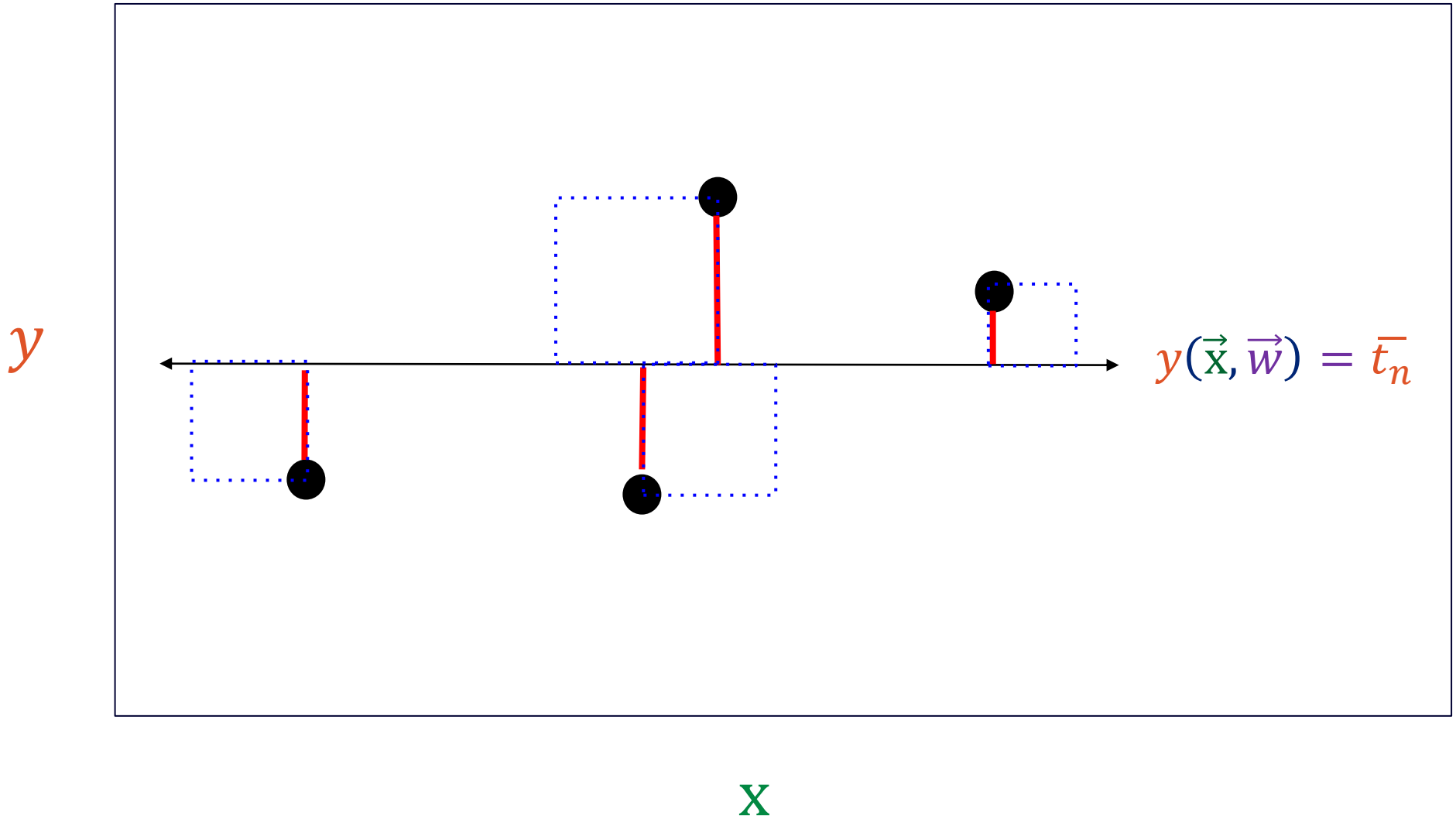
$$1 - \frac{\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2}{\sum_{n=1}^{N}(t_n - \bar{t_n})^2}$$

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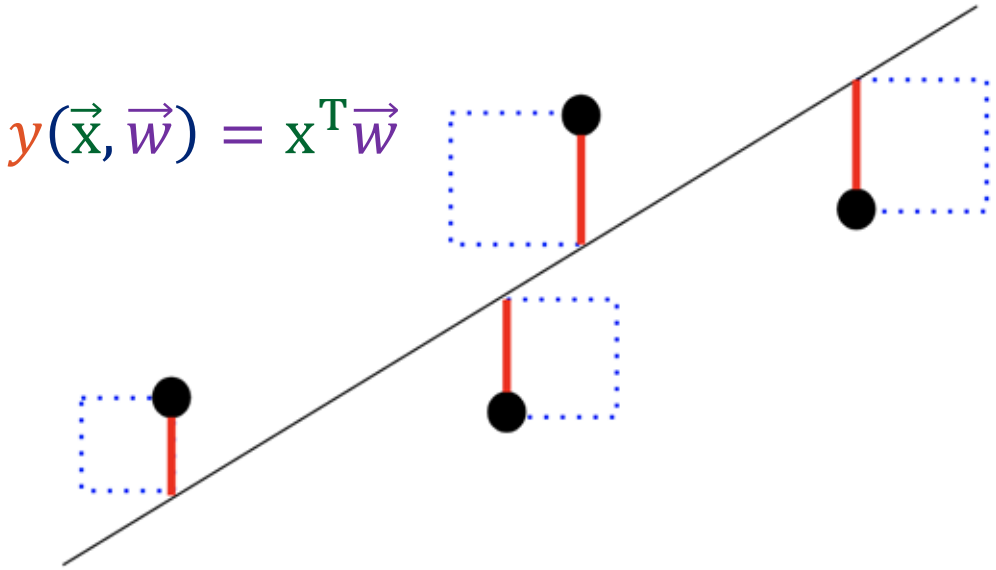
Visualizing the Sum of Squared Error of Regression Model

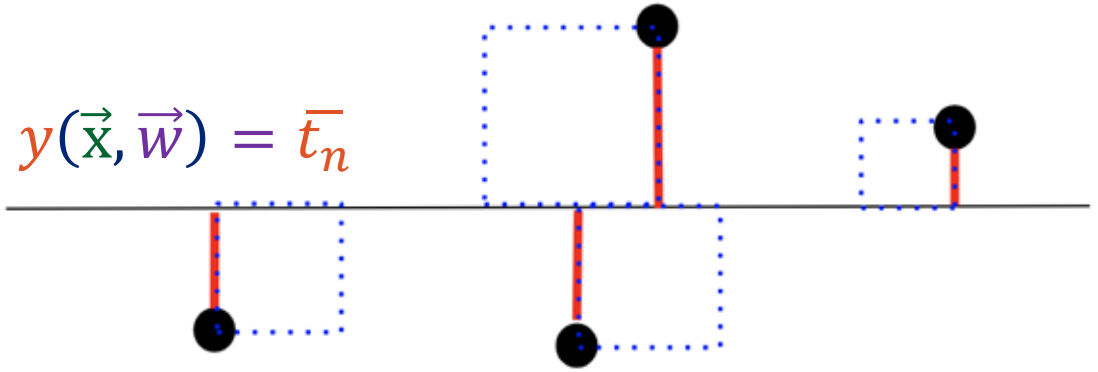


Visualizing the Sum of Squared Error of Intercept Model



R^2: Quality of the Fit Relative to Intercept Model





$$R^2 = \frac{\Delta \text{ in area}}{\text{Constant Model Area}} = 1 - \frac{\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2}{\sum_{n=1}^{N}(t_n - \overline{t_n})^2}$$

unitless and only compares performance relative to mean baseline.

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Evaluation - Metrics

Mean Squared Error (MSE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)$$

Root Mean Squared Error (RMSE)

$$\sqrt{\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)}$$

R-Squared (R^2) Score

$$1 - \frac{\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2}{\sum_{n=1}^{N}(t_n - \bar{t_n})^2}$$

Mean Absolute Error (MAE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}|t_n - y(x_n, w)|\right)$$

In the same unit as the data; similar to MSE but differs in how the penalization applies.

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Evaluation - Metrics

Mean Squared Error (MSE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)$$

Root Mean Squared Error (RMSE)

$$\sqrt{\frac{1}{N}\left(\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2\right)}$$

R-Squared (R^2) Score

$$1 - \frac{\sum_{n=1}^{N}\left(t_n - y(x_n, w)\right)^2}{\sum_{n=1}^{N}(t_n - \bar{t_n})^2}$$

Mean Absolute Error (MAE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}|t_n - y(x_n, w)|\right)$$

Mean Absolute Percentage Error (MAPE)

$$\frac{1}{N}\left(\sum_{n=1}^{N}\frac{t_n - y(x_n, w)}{t_n}\right) \times 100$$

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